

Fig. 3: Power Supply

of the bar graph voltmeter glows as pin 1 of IC1 is low and all other outputs pins are high. In this condition IC2(A) is enabled because input E (pin 1) is low. As inputs A0 and A1 of IC2(A) are high, its output Q3 goes low. This is Step 1 of step changer.

When voltage increases, input A0 of IC2(A) goes low and its output Q2 also goes low. This is Step 2 of step changer.

Both these outputs are combined with 1N4148 diodes and given to cathode pin of internal LED of IC7 (MOC3011). As internal LED of IC7 glows, TRIAC1 conducts and provides AC mains to tapping 'A' of mains transformer X2.

and output Q2 of IC2(A) generate enable input E for IC2(B) with the help of set and reset input pins (S and R) of flip-flop IC5(A) (CD4013). Pin 1 of IC5(A) provides low signal to enable input E of IC2(B) and output Q3 of IC2(B) goes low. This is Step 3 of step changer. Similarly, other conditions work in the same manner (see Table).

The number of tappings for transformer X2 and the number of solid-state relays to be used depend on the voltage range to be covered. If the minimum voltage can drop to 100 volts and the maximum could rise to 300 volts, we need to cover 200 volts deviation. This can be managed either through ten tappings with 20V difference or just five

When voltage increases further, both inputs A0 and A1 of IC2(A) go low, while both of its outputs go high, and TRIAC1 goes off. Input A1

PARTS LIST

Semiconductors:

- IC1 - LM3914 bar display driver
- IC2-IC4 - CD4556 dual decoder/demultiplexer
- IC5, IC6 - CD4013 dual flip-flop
- IC7-IC11 - MOC3011 optically isolated triac driver
- IC12 - 7812, 12V regulator
- TRIAC1 - BT136 triac
- TRIAC5 - 5A bridge rectifier
- BR1-BR5 - 1A bridge rectifier
- BR6 - 1N4148 switching diode
- D1-D10 - 1N4007 rectifier diode
- D11, D12 - 2 X 5 mm FLAT LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon unless stated otherwise):

- R1 - 5-kilo-ohm
- R2-R17, R23 - 1-kilo-ohm
- R18-R22 - 2-kilo-ohm
- VR1, VR2 - 10-kilo-ohm preset

Capacitors:

- C1-C8 - 100µF, 35V electrolytic
- C9 - 1000µF, 63V electrolytic
- C10 - 470µF, 35V electrolytic
- C11 - 0.1µF ceramic disk

Miscellaneous:

- X1 - 230V AC primary to 24V, 1A secondary transformer
- X2 - 120-280V AC (in 6 step), 1kVA stabilizer transformer (See text)
- F1 - 6A fuse

tappings with 40V difference between each.

Isolated solid-state power relay. Isolated solid-state power relay comprises an opto-isolator triac driver MOC3011, bridge rectifier (5A) and triac BT136. The opto-isolator triac driver MOC3011 is used for controlling the steps and connecting AC mains power supply to correct tapping of mains transformer X2 via solid-state relay. The capacity of solid-state relay depends on both the components—

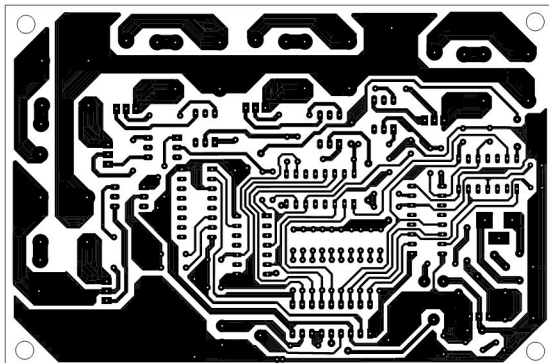


Fig. 4: Actual-size, single-side PCB for the Solid State voltage stabiliser